

Feature Zero: Who Fired First? Model-Agnostic Root-Cause Localization in ICS – A Case Study

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Abstract. Anomaly detectors for industrial control systems reliably report that something is wrong, but they say little about where the problem began. The operator is left to read a cascade of correlated alarms and guess the source. We treat that second step, root-cause localization, as a problem of its own and introduce Feature Zero, a model-agnostic framework for solving it. Given an anomaly window from any upstream detector together with a causal graph of the process, Feature Zero returns a ranked set of candidate root causes and nominates the single variable that most likely initiated the cascade. We call that earliest initiating variable the Feature Zero of the incident; it is a stricter notion than a root cause, since not every root cause started the cascade. We evaluate the framework as a case study on the Secure Water Treatment (SWaT) testbed and its six documented attacks. Using an expert causal graph built from the plant’s piping and instrumentation diagrams, Feature Zero ranks the true root cause first on five of the six attacks, outperforming multiple baselines.

Keywords: Root-cause analysis · Causal residuals · Temporal precedence · Changepoint detection · ICS security · Model agnostic

1 Introduction

Anomaly detection for industrial control systems (ICS) has matured. A plant can run a deep predictive model, a one-class detector, or a multivariate statistical monitor and receive an alarm within seconds of an incident. The step that has not kept pace is the one that comes next: once the alarm fires, telling the operator where the incident began.

The two tasks are not the same. A detector answers *when* something is wrong. To isolate the faulty component before the fault spreads, the operator needs to know *where* it started. In interconnected systems such as ICSs, one compromised variable affects others within seconds, so by the time the alarms go off dozens of channels are already abnormal. Common practice is to rank them by deviation from baseline, which finds the loudest signal rather than the source. Take a concrete case. An attacker pins a tank’s level sensor at a plausible value. The controller keeps the outlet pump running, the tank drains, and once the pump

pushes air its motor current climbs. When the detector fires, the pump is the loudest signal and the compromised sensor the quietest, so ranking by deviation places the pump first and the sensor deep in the list. The injection point is exactly what the attack was built to hide.

Ranking by anomaly score or reconstruction error, as most explainability and feature-importance methods do, rests on the same assumption: that the most anomalous-looking feature is the cause. The assumption is reasonable for hardware faults and noisy sensors. It is wrong for spoofing, command injection, and anything else built to avoid a magnitude signature at the injection point.

Feature Zero treats root-cause localization as a distinct problem, separate from detection. It takes an anomaly window from any detector and a causal graph of the system, then returns two outputs: a ranked list of plausible root causes, and a single nomination for the variable that most likely started the cascade, the Feature Zero.

Contributions.

- We propose Feature Zero, a model-agnostic framework for root-cause localization that works with any upstream anomaly detector.
- It identifies plausible root causes by combining parent-conditional residuals, statistical testing, onset estimation, and causal filtering.
- It nominates the *earliest* initiating anomaly, the Feature Zero, using temporal-precedence filtering and learned propagation delays.

The rest of this paper is organized as follows: Section 2 reviews previous work. Section 3 develops the method. Section 4 evaluates it on SWaT against seven baselines, with an ablation and a stability analysis. Section 5 concludes the paper.

2 Related Work

Locating the origin of an incident amid correlated alarms draws on four areas, covered below. Each addresses part of the problem, but none, to our knowledge, combines causal structure, temporal breaks in predictive residuals, and detector-agnostic design to identify the *earliest* broken mechanism.

Anomaly detection in ICS. Most operational ICS detectors are magnitude-first. Deep predictors and correlation-based methods flag large reconstruction errors or deviations from normal behavior [26,2]. These approaches work well when the root cause produces a clear amplitude change, but they assume the most abnormal signal is the cause. That assumption breaks on stealthy attacks such as sensor spoofing, where the compromised variable barely deviates. They detect *that* an anomaly occurred without reasoning about *which* mechanism failed first.

Explainable AI for anomalies. A parallel line of work applies post-hoc attribution methods such as SHapley Additive exPlanations (SHAP) or Local Interpretable Model-agnostic Explanations (LIME) [19,23] on top of deep detectors (for example, LSTM autoencoders) to highlight which sensors contributed most to an anomaly score [13,12]. SHAP and LIME return a feature ranking, but they rank contributions to a detector’s score rather than broken causal mechanisms. A sensor tracking a compromised parent still gets a large attribution because the detector reads it as anomalous, and the two rankings disagree precisely on the stealthy attacks we care about.

Changepoint detection. Penalized-cost methods such as PELT [16] efficiently detect distributional shifts in univariate signals and are commonly applied directly to raw process variables [18]. These approaches treat each channel independently and cannot distinguish a structural break in a causal mechanism from a downstream response. Feature Zero uses changepoint detection as a primitive, but applies it to parent-conditional residuals rather than raw signals, which makes it sensitive to broken causal relationships instead of large amplitude deviations.

Causal and graph-based root-cause analysis. Recent ICS work uses causal graphs to trace anomaly propagation. Several methods learn dynamic or multi-view causal graphs and detect attacks by comparing graph structure across time [21]. Causal Bayesian frameworks combine dynamic graph learning (for example, NOTEARS [28]) with Bayesian inference for root localization on SWaT data [9], or rank nodes by centrality and structural influence [14,5]. Closest to our setting, Ji et al. [15] and Rehak et al. [22] pair causal graphs with anomaly evidence for root-cause analysis. Both score the raw signals, though: they never check whether the parent-child mechanism itself broke, or which break came first.

These approaches give interpretable propagation paths, but they rank on correlations, topology, or aggregate scores rather than treating the *earliest* broken mechanism as a temporal target in a post-hoc, detector-agnostic setting. The baselines in Section 4 cover all three families discussed above: statistical (PCA contribution, VAR residuals, Random Forest, Isolation Forest, magnitude ranking), attribution (SHAP), and causal data-driven (GCAD, a Granger-causality ranker).

3 Methodology

We observe a multivariate time series $\mathbf{X}(t) = [X_1(t), \dots, X_d(t)]^\top$, where $d \in \mathbb{N}$ is the number of monitored variables, and are given two inputs: historical normal-operation data D_N and an anomaly window from any detector. The detector emits a set of suspect timesteps W ; the anomaly window is the data over those timesteps, $D_A = \{\mathbf{X}(t) : t \in W\}$. The two periods do not overlap. Given D_N and D_A , the goal is to find a set of plausible root causes and to select the earliest among them, which we call the Feature Zero. The two notions are not interchangeable: every Feature Zero is a root cause, but not every root cause

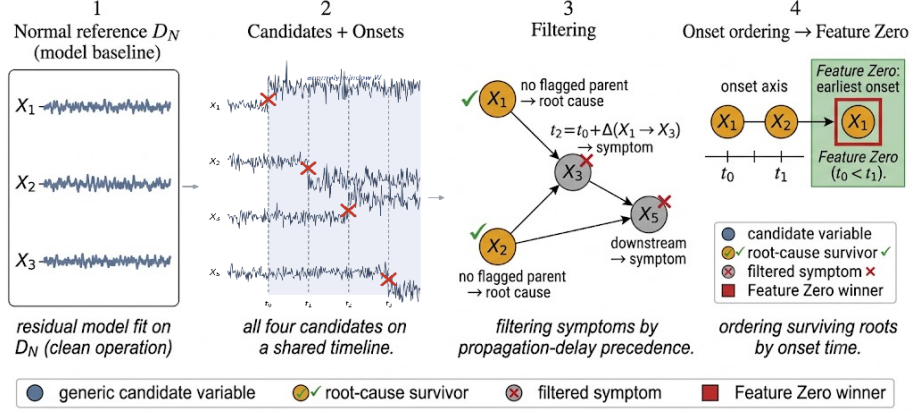


Fig. 1: Overview of Feature Zero: given a normal reference D_N , an anomaly window W , and a causal graph G , the framework returns the ranked root-cause candidates $\mathcal{C}$ and nominates X_0 . X_3 breaks one learned delay after X_2 , so the filter reads it as propagation and drops it. X_2 has no earlier break upstream to explain it, so it stays in $\mathcal{C}$. X_1 broke first, and is the nomination.

Algorithm 1 Feature Zero

Require: Normal reference D_N , anomaly window D_A , causal graph G

Ensure: Ranked root-cause candidates $\mathcal{C}$, earliest initiator X_0

- 1: Estimate delays $\Delta(j, i)$ for each edge via cross-correlation on D_N
 - 2: Fit OLS models $f_i(\text{Pa}(X_i))$ on D_N ; compute residuals $\mathbf{r}_N, \mathbf{r}_A$
 - 3: **for** each $X_i \in V$ **do**
 - 4: Test residuals for variance, mean, and correlation breaks
 - 5: Combine p-values (Fisher’s method); flag if significant after correction
 - 6: Estimate onset time o_i (mode change instead, if X_i is discrete)
 - 7: **end for**
 - 8: Drop flagged variables explained by an earlier ancestor’s break
 - 9: Score remaining candidates by statistical strength and downstream reach
 - 10: Cluster by onset; return ranked $\mathcal{C}$, with the top scorer in the earliest cluster as X_0
-

is a Feature Zero. In a coordinated attack with two injection points, both are root causes; the one that fired first, and from which the cascade can be traced, is the Feature Zero. Operators care about the full root-cause set to understand the incident, as well as the initiator to stop it.

3.1 Feature Zero framework

The framework follows four steps, from residual modeling, statistical evidence, and onset estimation to causal filtering. Details of the framework are given in Algorithm 1 and Figure 1 illustrates the pipeline.

3.2 Residual modeling

The framework needs a causal graph that specifies which variables directly fed into which others. Where there are P&ID documents from a plant, the graph can be extracted from them; otherwise, it can be learned. Section 3.7 covers how we built it for this paper. For each variable X_i we fit a linear parent-conditional model f_i on the normal reference set D_N with lag $\ell = 1$, standardizing the parent features:

$$X_i(t) = f_i(\text{Pa}(X_i)(t-1)) + \epsilon_i(t). \quad (1)$$

$\mathbf{r}_{N,i}$ denotes the residuals on D_N , and $\mathbf{r}_{A,i}$ those on the anomaly window D_A . Each f_i is fit once, by ordinary least squares on D_N alone, with parent values at $t-1$ standardized. No attack data or labels enter this step. A variable with no parents keeps its raw signal as its residual, so the tests below still apply to it.

Under normal operation f_i predicts X_i well from its parents, so $r_{A,i}$ stays close to $r_{N,i}$. When X_i is compromised the residual grows. A downstream X_j that merely follows a compromised parent still obeys its own f_j , so $r_{A,j}$ stays quiet. The residual points at the injection, not at the channels it disturbed.

3.3 Statistical evidence

For each variable we test whether its residuals have changed under the anomaly by comparing the normal and anomaly distributions. Let $\mathbf{p}_i = (p_{i,1}, \dots, p_{i,K})$ be a vector of p -values from K statistical tests on $\mathbf{r}_{N,i}$ versus $\mathbf{r}_{A,i}$. We combine them with Fisher’s method [7]:

$$p_{\text{comb},i} = 1 - F_{\chi^2_{2K}} \left(-2 \sum_{k=1}^K \log p_{i,k} \right), \quad (2)$$

where $F_{\chi^2_{2K}}$ is the CDF of the chi-squared distribution with $2K$ degrees of freedom. We then apply a Bonferroni correction [6] over the d variables, and flag a variable if $p_{\text{comb},i} < \alpha/d$. Thresholds are bootstrapped from D_N so that they adapt to each sensor’s noise instead of being fixed globally. In the SWaT case study we use $K = 3$ tests: a variance-ratio test for increased volatility, a mean-shift test for a systematic offset, and a correlation-break test for a failure of the parent-child relationship itself.

3.4 Onset estimation

For each flagged variable we estimate the onset time o_i as the earliest of the changepoint times returned by two detectors run on the centered residuals: a rolling-window exceedance detector and a CUSUM detector for cumulative drift. For low-cardinality discrete variables such as binary pump and valve states we add a third, mode-change detection, and take the earliest of the three. The onsets drive the precedence filter.

3.5 Causal precedence filter

This is the step that turns a set of flagged variables into an ordered set of root causes and an initiator. For each edge $X_j \rightarrow X_i$ we estimate the expected propagation delay $\Delta(j, i)$ from D_N by maximizing the absolute cross-correlation of the centered parent and child series over non-positive lags. A flagged node X_v with onset o_v is removed from the candidate set if some flagged ancestor X_a meets two conditions: $o_a \leq o_v + \delta$, where δ is a fixed grace tolerance (Table 1) that absorbs onset-estimation error, and $o_v - o_a \geq \Delta(a, v) - \tau$ for a delay tolerance τ .

The second condition is one-sided by design. It rules out a descendant whose break arrives sooner than the edge could have carried the change, but it does not bound how late a response may be, so a break that lags far behind its expected delay is still treated as propagation. A downstream break is discarded only when an earlier upstream break can account for it. Everything else stays in the candidate set as a genuine root cause.

3.6 Root causes and Feature Zero selection

All surviving candidates are scored by the strength of their combined statistical evidence and their downstream reach: $s_r = -\log_{10} \max(p_{\text{comb}, r}, 10^{-100})$ is the negative log of the Fisher-combined p -value from Eq. (2), capped to avoid numerical artifacts from vanishing p -values, and ρ_r is the fraction of r 's descendants in G that are flagged. The final score is $s_r(1 + 0.5\rho_r)$, so a candidate whose effects spread widely scores higher. Candidates are ordered by onset first and score second, with nearby onsets clustered so a coordinated multi-point attack is reported as one event. Candidates with distant onsets fall into separate clusters, and all clusters stay in the output. The full ordered set of root-cause candidates is $\mathcal{C}$; the top-scoring member of the earliest cluster is nominated as the Feature Zero.

Feature Zero is model-agnostic: it needs only the anomaly window W from the upstream detector. It never reads the detector's internal scores or gradients, and it trains its residual models independently on D_N .

3.7 The Causal Graph

A directed causal graph $G = (V, E)$ is required because timing alone cannot differentiate causes from effects: when a sensor fault triggers downstream alarms, both the sensor and the affected devices turn abnormal within seconds, and both look equally culpable. Conditioning each variable on its parents breaks the tie.

The causal graph can come from two sources. In engineered systems, process structure is already documented in piping and instrumentation diagrams (P&IDs) and control logic; these can be translated directly into a directed acyclic graph encoding dependencies. When documentation is unavailable, causal structure can be learned from normal-operation data using methods like Granger

causality [11] or graphical lasso [8]. Learned graphs are less accurate than expert-elicited ones due to identifiability constraints, but they are useful when documentation does not exist. Feedback control loops can make the static wiring cyclic, but every edge acts with a lag of at least one timestep, so the time-unrolled graph is acyclic by construction and the method stays well-defined. A cycle survives unrolling only when the plant is sampled more slowly than its control loops react. The precedence filter would then have to run on the condensation of the graph, with onset alone ordering the variables inside each strongly connected component. Our expert graph has no such cycle, so we leave that case for future work.

3.8 Threat model and assumptions

We assume an adversary who can alter sensor readings and actuator commands while the plant runs, by compromising a PLC or by tampering with the field network. The attacker is free to choose values that look plausible on the compromised channel, which covers the six SWaT attacks and the stealthy cases the framework is built for.

Two things are assumed out of reach. The adversary cannot rewrite the historian record of the normal period, and cannot alter the P&ID or control description the graph is read from. Both assumptions are load-bearing, and neither is guaranteed in practice. Section 5 sets out what the framework does when they fail.

4 Case Study: Secure Water Treatment Testbed (SWaT)

We validate the framework on the SWaT benchmark [10], which is well documented and has full ground truth for six attack campaigns with recorded timestamps. The goal is to show that the framework recovers documented root causes on a real system and to find out which components matter. We claim no general superiority: this is one realistic ICS scenario.

4.1 Dataset

The Secure Water Treatment testbed is a fully operational six-stage water-purification plant with 51 sensors and actuators. We use the public 2019 attack collection: 10,280 timesteps of normal operation followed by six documented attacks. The attacks are two sensor spoofs (FIT401 held at 0.5; LIT301 held at 1024), three actuator-command injections (P601 forced on, MV501 forced closed, P301 forced off), and one multi-point attack (MV201 forced open with P101 forced on). We use the documented attack windows as the anomaly windows W , which simulates a perfect upstream detector and isolates root-cause performance from detection performance. In deployment W would come from a real detector and inherit its noise, which we probe in Section 4.3.

Table 1: Parameters for the SWaT experiments. Counts in samples are timesteps at one-second resolution. Plant-sourced values are read from the process timescale; fixed values were set once, and Section 4.3 shows the result is insensitive to their exact values.

Parameter	Symbol	Value	Source	Role
Model lag	ℓ	1	plant	Parent values at $t - 1$ predict the child at t
Significance level	α	0.05	conv.	Bonferroni-corrected over the d monitored variables
Adaptive thresholds		200 / 95–96 / 2.0–2.5	conv./fixed	Bootstrap resamples (conv.); percentiles and floors for variance and mean-shift tests (fixed)
Onset detection		3 of 5 / 2.5	fixed	Rolling exceedances required, and CUSUM threshold (drift 0.5)
Causal grace	δ	18	plant	Slack when ordering an ancestor against a descendant
Propagation delay	τ	30 / 2	plant	Cross-correlation search range, and slack against $\Delta(j, i)$
Onset cluster gap		3	plant	Onsets within this span form one event

4.2 Implementation and parameters

Table 1 lists every parameter of the Feature Zero pipeline used in the SWaT runs. The residual models are ordinary least squares, and nothing else in the pipeline is fit.

The plant-sourced parameters are read from the process rather than tuned: the model lag follows from one-second sampling, and the causal grace, delay search range, and cluster gap come from settling times measured in the normal period, meaning how long a controlled variable takes to react to an upstream change and stabilize. That interval bounds a plausible cross-stage propagation delay and sets the search window. For the fixed parameters we claim no optimality; they were set once, and Section 4.3 shows the floors can be cut to a fifth of these values without changing which variable Feature Zero nominates. Computational cost is linear in d : $O(d\bar{p}^2n_N)$ to fit the regressions once on D_N , plus $O(dn_A)$ per incident window.

4.3 Experiments and Results

Evaluation metrics. All experiments report top- k accuracy (the fraction of attacks where the true root cause is ranked in the top k), the mean rank of the true root cause, and mean reciprocal rank (MRR); top- k and MRR are standard in root-cause analysis [24]. We call top-1 accuracy the Feature Zero Identification Rate (FZIR) and use that name in all tables. A method that does not return the true root cause at all is assigned a sentinel rank of 99, so mean ranks above $d = 51$ in the tables below reflect misses, not positions in a ranking.

Comparison of graph sources We tested Feature Zero with three graph sources: (i) an expert graph manually extracted from the plant’s P&ID; (ii) a

Table 2: Feature Zero performance with different causal graph sources on SWaT (six attacks).

Graph source	FZIR	Top-3	Top-5	Mean rank	MRR
Expert (P&ID)	83%	83%	83%	3.0	0.846
DAGL (data-driven)	33%	50%	50%	46.5	0.401
Granger (data-driven)	33%	33%	33%	79.7	0.339

Granger-Lasso graph learned from the normal period using Vector Autoregressive (VAR) regression with L1 penalty [1]; and (iii) a Directed Graphical Lasso (DAGL) graph from sparse inverse covariance estimation on the same period [8]. None of the learned graphs saw attack labels. To build the expert graph we read the sensor and actuator table and process overview published with the dataset [10], adding an edge where the documentation indicated a direct dependency and extending the same patterns to instruments the 2015 table does not cover.

The expert graph reaches 83% FZIR, five of the six attacks, while both data-driven graphs stop at 33% (Table 2). The learned graphs fail on sensor spoofs, where the causal structure is subtle and linear lag assumptions break. An expert graph wins when one exists; Granger-Lasso or DAGL are usable fallbacks otherwise, at markedly lower accuracy.

Comparison against baselines. We compare Feature Zero with seven baselines from three families: statistical, attribution, and causal methods, the same families surveyed in Section 2. GCAD stands in for the causal line: the causal Bayesian and graph-centrality frameworks evaluated on SWaT [9,14,5] perform detection and localization jointly and release no implementation, so GCAD is the closest causal method we could run on identical windows.

Table 3 shows that Feature Zero recovers the true root cause on five out of six attacks (83% FZIR), against 33% for the strongest baseline (PCA). The gap comes down to one design choice. Baselines rank variables by how unusual the raw signal looks, and on a stealthy attack that ordering runs backwards: the compromised channel barely moves while the actuators chasing it swing hardest. How well a baseline finds the compromised channel depends on how far the held value sits from the sensor’s normal range: when it stays in range, no baseline ranks it better than fifth (Table 4). Conditioning on parents needs no such signature: the frozen sensor stops agreeing with what its parents predict, while the actuators responding to it still match their own parents to within noise. Figure 3 shows both spoofs this way.

Table 4 breaks the aggregate scores down by attack. Feature Zero places the documented root first on five of the six. The exception is A1: FIT401’s residual break falls outside the window SWaT 2019 labels as the attack, so the ranking never sees the evidence (Section 4.3). Isolation Forest’s rank of 5 there is the one case where a baseline beats us. No baseline is first on more than two attacks,

Table 3: Root-cause ranking on SWaT (six attacks): Feature Zero with the expert graph against seven baselines on identical anomaly windows. Bold indicates the best value in each metric.

Method	FZIR	Top-3	Top-5	Top-10	Mean rank	MRR
Feature Zero	83%	83%	83%	83%	3.0	0.846
<i>Heuristic / Statistical</i>						
PCA contribution [25]	33%	33%	50%	50%	12.8	0.395
VAR residuals [20]	17%	17%	33%	33%	18.3	0.239
Random Forest [3]	17%	17%	33%	33%	13.3	0.238
Isolation Forest [17]	0%	17%	33%	50%	26.3	0.129
Magnitude [27]	17%	50%	67%	83%	5.3	0.355
<i>Attribution</i>						
Attribution (SHAP) [19]	0%	0%	17%	33%	15.7	0.091
<i>Causal data-driven</i>						
GCAD (Granger causality) [4]	17%	33%	50%	50%	28.0	0.293

Table 4: Rank of the documented root cause on each SWaT attack. Feature Zero uses the expert graph. Rank is the position of the true root in each method’s output; for Feature Zero a root sharing the earliest estimated onset takes rank 1, since simultaneous onsets carry no temporal ordering. For the multi-point attack A4 the rank of the earlier injection point (MV201) is reported. Bold marks the best rank on each attack.

	A1	A2	A3	A4	A5	A6
Method	FIT401	LIT301	P601	MV201	MV501	P301
Feature Zero	13	1	1	1	1	1
PCA contribution	18	1	41	11	1	5
VAR residuals	31	40	1	21	4	13
Random Forest	20	1	5	17	24	13
Isolation Forest	5	7	66	3	62	15
Magnitude	14	7	1	4	3	3
Attribution (SHAP)	24	7	5	17	25	16
GCAD	54	42	64	2	5	1

and each fails somewhere different. PCA finds LIT301 and MV501 but puts P601 41st. VAR recovers P601 and misses LIT301 by forty places. GCAD does best on P301 and MV201, both actuator attacks, but ranks the two spoofed sensors 54th and 42nd. Magnitude ranking has the best baseline mean rank here, 5.3 against 12.8 for PCA, yet reaches first place only once. That is what the aggregates hide: getting close to the source is easier than identifying it.

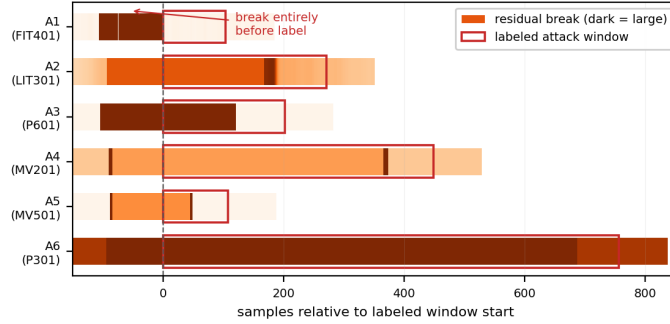


Fig. 2: Residual evidence versus dataset labels for all six SWaT attacks. Each strip shows the parent-conditional residual magnitude of the documented root-cause sensor, normalized per attack so that dark marks its break. The red outline is the window labeled as the attack in SWaT 2019.

A labeling offset in SWaT 2019 Assembling the residual evidence turned up a property of the dataset that affects any method evaluated on the labeled window boundaries. We scanned each root-cause variable for the interval where it sits pinned at its spoofed or forced value and compared that against the window SWaT 2019 labels as the attack. The labels run 80–100 samples late on all six attacks (Figure 2): the physical event and the label do not describe the same interval. The physical fault begins when the attacker starts pinning the sensor, but the plant is slow and the labels seem to have been placed once the downstream effects were visible. The label interval is a suffix of the true event, shorter by about the settling time. That suffix still overlaps the residual break on five attacks, so the tests fire and the root ranks first. A1 is the exception: the injection ends two samples before the label begins, so Feature Zero sees a window in which the sensor already reads normally (rank 13). Widening the A1 window 30 samples earlier moves the sensor to rank one. We report the as-labeled result for comparability with prior SWaT work.

What a break looks like. Figure 3 shows the two sensor spoofs side by side. In both, the residual opens a sustained offset that the raw trace does not show, which is the evidence the mean-shift test acts on. The FIT401 panels also make the labeling offset visible: its residual break sits almost entirely outside the red label outline, whereas LIT301’s runs the length of it.

The multi-point attack. The multi-point attack (MV201 and P101) creates two genuine root causes. Both are detected and land in the same earliest-onset cluster (onset = 0), so the coordinated event is recovered as a single cluster. Within that cluster, however, MV302 and P301 outscore the attack points, and the nomination goes to MV302; P101 and MV201 rank third and fourth in $\mathcal{C}$.

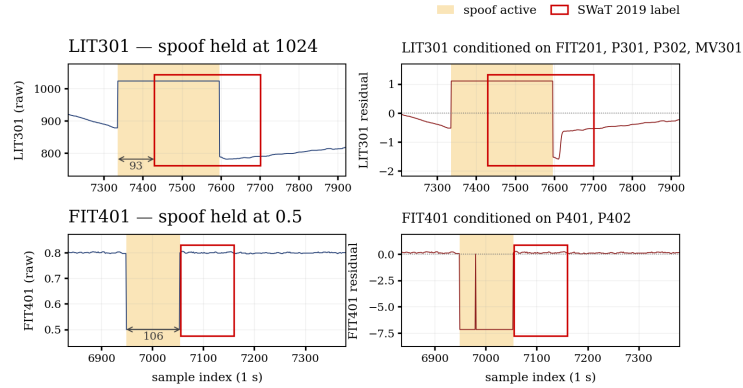


Fig. 3: The two SWaT sensor spoofs. Top row: LIT301. Bottom row: FIT401. Left column shows the raw sensor trace, right column the parent-conditional residual. The shaded band is the interval over which the sensor is pinned at its spoofed value; the red outline is the window labeled as the attack in SWaT 2019. In both cases the raw trace flattens while the residual breaks, and the residual break precedes the label.

Table 5: Ablation on SWaT with the expert graph.

Configuration	FZIR	Top-3	Top-5	Mean rank	MRR
Full Feature Zero	83%	83%	83%	3.0	0.846
No statistical tests	0%	0%	0%	99.0	0.010
No onset ordering	83%	83%	83%	5.0	0.840
No discrete handling	67%	67%	83%	3.5	0.721

The onset-tie metric credits this as a rank-1 hit, but by strict nomination the top pick is a bystander: the honest limit of scoring by residual magnitude when several variables break at once.

Ablation study. Table 5 reports the effect of removing each of Feature Zero’s components in turn: the residual tests, onset-based precedence, and mode-change detection for binary actuators. The residual tests are essential: disabling them collapses FZIR to 0%. Removing onset ordering leaves FZIR unchanged but raises mean rank from 3.0 to 5.0. Turning off discrete handling costs one attack, the binary-actuator case, and drops FZIR to 67%. Most of the signal comes from the combined p-value of the variance and mean tests; the other components add robustness and consistent ranking, not FZIR itself.

Stability Analysis. A real detector will not hand over a clean window, so we tested two kinds of perturbation, each repeated over 10 random seeds (Figure 4).

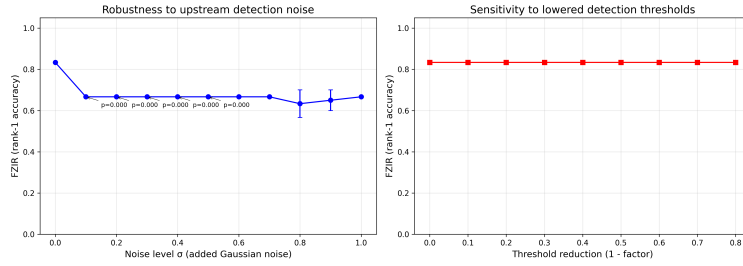


Fig. 4: Stability of Feature Zero (expert graph) on SWaT. (left) Mean FZIR with one standard deviation against added Gaussian window noise σ . (right) Mean FZIR against the threshold-reduction factor f , where $f = 1$ is the default and $f = 0.2$ is 20% of the default thresholds. Error bars are one standard deviation over 10 random seeds.

The first adds Gaussian noise with standard deviation σ from 0 to 1.0 to every point in each attack window. This perturbs values inside the window, not its boundaries, so it stands in for a noisy or uncleaned window rather than a misplaced one; that cost is a separate question, which Section 4.3 answers for the one attack where the boundary is demonstrably wrong. FZIR starts at 83% and settles around 67% under moderate noise ($\sigma \leq 0.5$), never dropping below 63% even at $\sigma = 1.0$, with a standard deviation across seeds of at most 0.067. With six attacks, that drop is one extra miss, and 67% under noise is still double what the best baseline manages on clean windows.

Second, we lowered the variance-ratio and mean-shift floors to as little as 20% of their defaults, simulating an over-sensitive configuration. FZIR stayed at 83% throughout, and a paired t -test found no significant change ($p > 0.05$). The default thresholds are therefore not finely tuned, and it is the adaptive tests, not the fixed floors, that do the work.

4.4 Summary of findings

The experiments support four findings. (1) Graph quality is the binding constraint. Replacing the expert graph with a learned one cuts FZIR from 83% to 33% (Table 2), and the framework has no way to notice the damage: a variable conditioned on parents that do not drive it produces a large residual for the wrong reason, and that residual is indistinguishable from a genuine break. (2) The residual statistical tests carry the result. Disabling them alone collapses FZIR to 0%, while removing precedence filtering or discrete-variable handling moves mean rank or one attack class but leaves rank-1 accuracy intact (Table 5). (3) The precedence filter contributed little on SWaT. Cascades cross a stage in seconds, so the tests had usually isolated the source before the filter ran; on plants with slower dynamics, where downstream breaks trail the source by minutes rather than seconds, we expect it to matter more. (4) The single miss is

a labeling artifact, not a method failure. The A1 label window opens after the short injection has already ended, so the method is scored on a window that contains no attack signal (Section 4.3).

5 Conclusion

Anomaly detectors in industrial control systems report that something is wrong but not where it began, and on stealthy attacks the loudest channel is rarely the injection point. We built Feature Zero to close that gap: given an anomaly window from any upstream detector and a directed causal graph of the process, it fits parent-conditional residual models on normal data, tests each residual for a distributional break, orders the survivors by estimated onset, and filters out breaks an earlier ancestor already explains. The output is a ranked set of root-cause candidates $\mathcal{C}$ with the earliest initiator nominated as the Feature Zero of the incident.

The case study on SWaT 2019 shows this design works and shows why it works. Against seven baselines, conditioning on causal parents rather than reading raw signatures is what recovers stealthy attacks: Feature Zero ranks the true root cause first on five of six attacks (83% FZIR) against 33% for the best baseline, and the margin survives window noise up to $\sigma = 1.0$ and detection floors cut to a fifth of their values. That margin traces to the residual tests and to the causal graph itself, and the one miss traces to a scoring artifact rather than a method failure. How well the graph captures the plant is what sets the ceiling on localization quality; the residual machinery built on top of it matters less.

This is a proof of concept, not a general guarantee. Feature Zero’s one load-bearing assumption is that a compromised variable stops agreeing with its causal parents, and the case study also delimits where that assumption fails. An adversary who spoofs a sensor and its parents in a mutually consistent way leaves the parent-child relationship intact, and the incident stays invisible even as the plant is driven somewhere unsafe. A root cause outside V , such as a failed power supply or a compromised engineering workstation, reaches the process only through the variables it disturbs, so the nomination is a symptom. A wrong graph fails symmetrically: a missing edge leaves a broken variable with no ancestor to explain it and promotes a symptom to root cause, while a spurious edge provides an ancestor whose ordinary behavior appears to explain a real break and filters the true source out as propagation. A reference that no longer describes the plant breaks everything downstream of it, whether through drift or through a poisoned D_N , and that failure is silent, which makes periodic revalidation of D_N the safeguard we would recommend first. Linear residual models leave structured residuals on saturating actuators and switching control, and we have not measured what that costs. Six attacks on one testbed is a case study, not a benchmark.

These limits set the next steps. The graph-quality finding wants testing on additional ICS benchmarks to see whether it generalizes, and the saturating and switching cases call for nonlinear residual models in place of the linear ones used here. Hybrid graphs that seed a learned structure with whatever P&ID exists

would also help close the gap we found between the expert and learned conditions. We also want to pair Feature Zero with several upstream detectors, since model-agnosticism is so far a claim about the interface rather than a measured result, and to run it on windows produced by a real detector rather than ground truth. Bounding propagation delays from above would let the precedence filter discard late responders as well as breaks arriving too early. Plants sampled too slowly to unroll their control loops need a precedence filter that works on the condensation of the graph, and we have not built one.

Availability. The implementation, the expert graph, and the experiment configurations are available from the authors upon request.

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